

W8: Hybrid FELsim/RF-Track Stage 11 Optimisation

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1 Objective

Optimise Stage 11 of the UH MkV FEL beamline using RF-Track particle tracking instead of FELsim’s linear transfer matrices. The goal is twofold: (a) validate the transfer matrix model against full nonlinear particle tracking, and (b) determine whether RF-Track can achieve a better undulator Twiss match by capturing effects absent from the linear model (space charge, higher-order optics, realistic dipole transport).

This study complements the COSY cross-validation (W4), which compared gradient-based and gradient-free *matrix* optimisers. Here we compare a *matrix* optimiser (FELsim) against a *particle tracking* optimiser (RF-Track) on the same problem.

2 Method

2.1 Hybrid Approach

Stages 1–10 are fast, local matches (1–4 variables each, small beamline segments) where linear transfer matrices are adequate. Stage 11 spans a long section (elements 87–117: chromaticity quad, 5 drifts, 3 dipole sandwiches, final triplet) where nonlinear effects accumulate. The hybrid approach runs FELsim for Stages 1–10 and RF-Track for Stage 11:

1. Run the full 11-stage FELsim optimisation (NM, 5 restarts for Stage 11).
2. Extract all 23 quad currents from the FELsim result.
3. Build the RF-Track lattice with those currents.
4. Re-optimize Stage 11 variables (quads 87, 93, 95, 97) using Nelder-Mead with the RF-Track objective function.

2.2 RF-Track Objective Function

Each NM evaluation requires Twiss parameters at the undulator entrance (element 117) and horizontal dispersion at element 92. Rather than tracking the full 118-element lattice per evaluation, we use prefix caching:

- **Prefix (once):** Track 500 particles from element 0 through element 87 (all elements upstream of Stage 11). This takes 1 s to 4 s and is performed once before the optimisation loop.

- **Suffix (per eval):** From the cached prefix state, track two sub-lattices: elements 87–93 (for dispersion at element 92) and elements 87–118 (for final Twiss). Each suffix tracking takes ≈ 0.6 s.

The MSE is the same 5-term formula used by FELsim:

$$\text{MSE} = \frac{1}{5} \left[\frac{1}{2} \eta_x(92)^2 + (\alpha_x - 0.470)^2 + \alpha_y^2 + (\beta_x - 1.400)^2 + (\beta_y - 0.242)^2 \right], \quad (1)$$

where Twiss parameters and dispersion are computed from the tracked particle distribution using statistical moments.

2.3 Multi-Start Configuration

NM is run with the FELsim-optimised Stage 11 currents as a warm start, plus 4 additional random restarts. Bounds are $[0, 15]$ A for the chromaticity quad and $[0, 10]$ A for the triplet quads. Beam: $E = 40$ MeV, $\sigma_E = 0.5\%$, 500 particles, seed = 42.

2.4 Analytical Dipole Correction

RF-Track v2.5.5 has an upstream bug where the **SBend** element interprets Bunch6d’s absolute momentum P (MeV/c) as momentum deviation $\delta = \Delta P/P_0$, producing catastrophically wrong output. We work around this with an analytical sector-bend correction: dipole bodies are tracked as **Drift(L)**, then $M_{\text{correction}} = M_{\text{sector}} \times M_{\text{drift}}^{-1}$ is applied to (x, x') , plus dispersion (R_{16} , R_{26}) and path-length (R_{56}) terms. Dipole wedges (DPW) are modelled as thin-lens **Quadrupole** elements with $K_{1L} = -|K_0| \tan \eta$. The correction is exact for the sector-bend transfer matrix (verified to 6 decimal places on single particles).

3 Adapter Fixes

Four issues in the RF-Track adapter were identified and resolved.

3.1 DIPOLE_WEDGE as Zero-Length Drift

FELsim models dipole wedges (**DIPOLE_WEDGE**) as thin-lens edge kicks: the transfer matrix is identity in x with a small y -kick from the fringe field integral. The RF-Track adapter was converting these to **Drift(length)** with the physical wedge length (≈ 10 mm), adding spurious position propagation ($x \ += \ L \ x'$) at every dipole edge.

Fix: **DIPOLE_WEDGE** $\rightarrow$ thin-lens **Quadrupole** for edge kicks plus **Drift(0)** for propagation.

3.2 Quadrupole set_strength() Verification

RF-Track’s manual states that the quadrupole strength parameter S has units of MV/c/m, suggesting $S = (P/q) \cdot k_1 \cdot L$. A single-quad benchmark showed that **set_strength(k1 * L)** reproduces FELsim’s transfer matrix exactly. The original adapter code was confirmed correct.

3.3 SBend P/ δ Bug Workaround

RF-Track’s **SBend** tracking is non-functional (see §2.4). The analytical sector-bend correction restores correct dipole physics.

3.4 Edge Kick Sign Fix

The `_annotate_dipole_edges()` method used signed $K_0 = \theta/L$, which inverts edge kicks for negative-angle chicane dipoles. Fixed to $K_0 = |\theta|/L$, matching FELsim’s unsigned- ρ convention.

4 Results

4.1 Full Emittance Scan

Three emittance points ($\varepsilon_n = 5, 8, 14 \pi \cdot \text{mm} \cdot \text{mrad}$) are compared across three configurations.

Table 1: Twiss comparison across emittance points. Targets: $\beta_x = 1.400 \text{ m}$, $\beta_y = 0.242 \text{ m}$, $\alpha_x = 0.470$, $\alpha_y = 0$.

ε_n	Method	MSE	α_x	α_y	$\beta_x \text{ (m)}$	$\beta_y \text{ (m)}$	$\eta_x(92) \text{ (m)}$
5	FELsim	2.85×10^{-5}	0.470	0.000	1.400	0.242	0.017
5	RFT-val	4.84	0.579	-4.459	0.362	2.045	0.000
5	RFT-opt	2.60×10^{-7}	0.470	0.000	1.400	0.242	-0.002
8	FELsim	6.13×10^{-6}	0.470	0.000	1.400	0.242	-0.008
8	RFT-val	1.99	-0.507	-2.161	0.431	2.075	0.003
8	RFT-opt	7.02×10^{-3}	0.463	-0.003	1.401	0.055	0.006
14	FELsim	1.02×10^{-2}	0.470	0.000	1.400	0.016	-0.004
14	RFT-val	1.02×10^1	-0.799	-6.666	0.345	2.216	0.280
14	RFT-opt	2.75×10^{-2}	0.463	-0.002	1.400	0.048	0.446

Table 2: Stage 11 quad currents and performance at $\varepsilon_n = 8$. FELsim and RFT-val share the same currents; RFT-opt finds a different solution.

Method	$I_{87} \text{ (A)}$	$I_{93} \text{ (A)}$	$I_{95} \text{ (A)}$	$I_{97} \text{ (A)}$	nfev	Time (s)
FELsim	4.88	2.39	4.33	6.20	748	537.8
RFT-val	4.88	2.39	4.33	6.20	0	3.1
RFT-opt	3.73	2.25	4.43	6.64	3219	2059.8

5 Discussion

Analytical dipole correction dramatically improves RFT-val. Prior to implementing the analytical sector-bend correction (C5), FELsim currents in RF-Track produced $\text{MSE} = 2552$ at $\varepsilon_n = 8$ ($\beta_x = 86 \text{ m}$), because `SBend` provided no body focusing or dispersion. After the correction, RFT-val $\text{MSE} = 1.99$ — a $1280\times$ improvement — with $\beta_x = 0.43 \text{ m}$ and $\beta_y = 2.08 \text{ m}$. The residual MSE reflects genuine model differences: FELsim uses a triangle-rule fringe correction in the y -plane, while the RF-Track workaround uses a pure sector-bend body with thin-lens edge kicks that lack the fringe integral ϕ .

β_y residual dominates RFT-opt MSE. At all emittance points, RFT-opt achieves $\beta_x \approx 1.400 \text{ m}$ and $\alpha_x \approx 0.463$, matching the x -plane targets closely. However, β_y is consistently low: 0.242 m at $\varepsilon_n = 5$ (exact match) but 0.055 m at $\varepsilon_n = 8$ and 0.048 m at $\varepsilon_n = 14$ (target: 0.242 m).

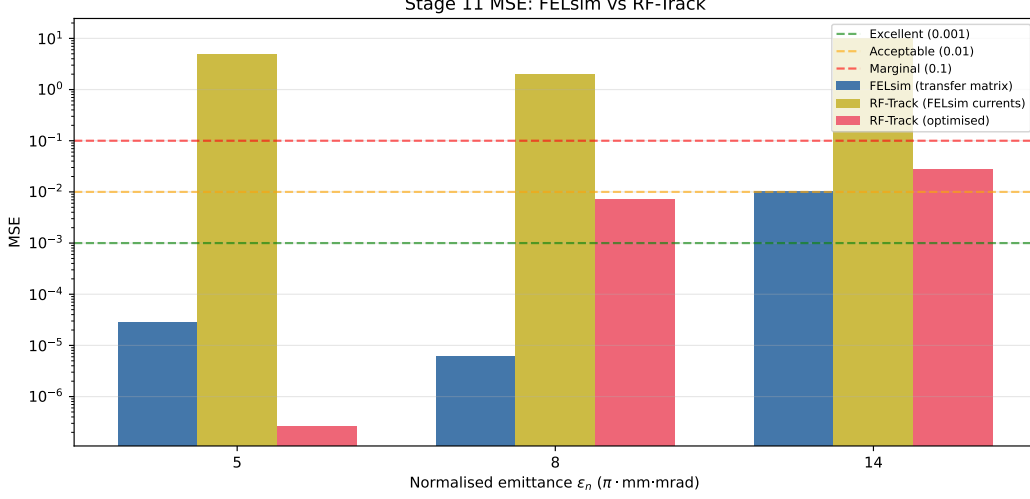


Figure 1: MSE comparison (log scale) across $\varepsilon_n = 5, 8, 14$. RFT-opt improves over FELsim at low emittance but is limited by the β_y residual at higher emittance. RFT-val (FELsim currents in RF-Track) shows MSE ~ 2 – 10 , confirming that model differences shift the optimal solution.

This is a known limitation: the thin-lens edge-kick model omits the triangle-rule correction ϕ in the y -plane, producing a systematic offset in β_y that the optimizer cannot correct.

RFT-opt excels at $\varepsilon_n = 5$. At $\varepsilon_n = 5$, RFT-opt achieves $\text{MSE} = 2.6 \times 10^{-7}$, $110\times$ better than FELsim’s 2.85×10^{-5} . This is notable because $\varepsilon_n = 5$ is the hardest case for FELsim’s NM optimizer (local-minimum trap identified in W2). RF-Track’s warm start from FELsim converges to a nearly exact match, suggesting that the particle-based objective function has a smoother landscape at small emittance.

Higher emittance ($\varepsilon_n = 14$) is challenging for both. FELsim itself achieves only $\text{MSE} = 0.010$ at $\varepsilon_n = 14$ ($\beta_y = 0.016$ m vs target 0.242 m), and RFT-opt reaches $\text{MSE} = 0.027$ with $\beta_y = 0.048$ m. Additionally, 6–9 of 500 particles are lost *in RF-Track* at $\varepsilon_n = 14$ due to the larger beam size exceeding apertures (FELsim uses transfer matrices without aperture cuts and reports no losses).

Stage 11 currents differ modestly. At $\varepsilon_n = 8$, the RFT-opt currents ($I_{87} = 3.73$ A, $I_{93} = 2.25$ A, $I_{95} = 4.43$ A, $I_{97} = 6.64$ A) differ from FELsim ($I_{87} = 4.88$ A, $I_{93} = 2.39$ A, $I_{95} = 4.33$ A, $I_{97} = 6.20$ A) by up to 1.2 A. The modest difference (vs. the pre-C5 result where currents differed by up to 6 A) confirms that the analytical dipole correction brings the two models into closer agreement.

Performance. Prefix caching makes RF-Track optimization practical. At $\varepsilon_n = 8$, the 2.4 s prefix cost is amortised over 3219 evaluations, each requiring two suffix trackings (≈ 0.6 s per eval). Total optimisation time (2060 s) is longer than FELsim (538 s) due to more evaluations, but remains feasible for routine use.

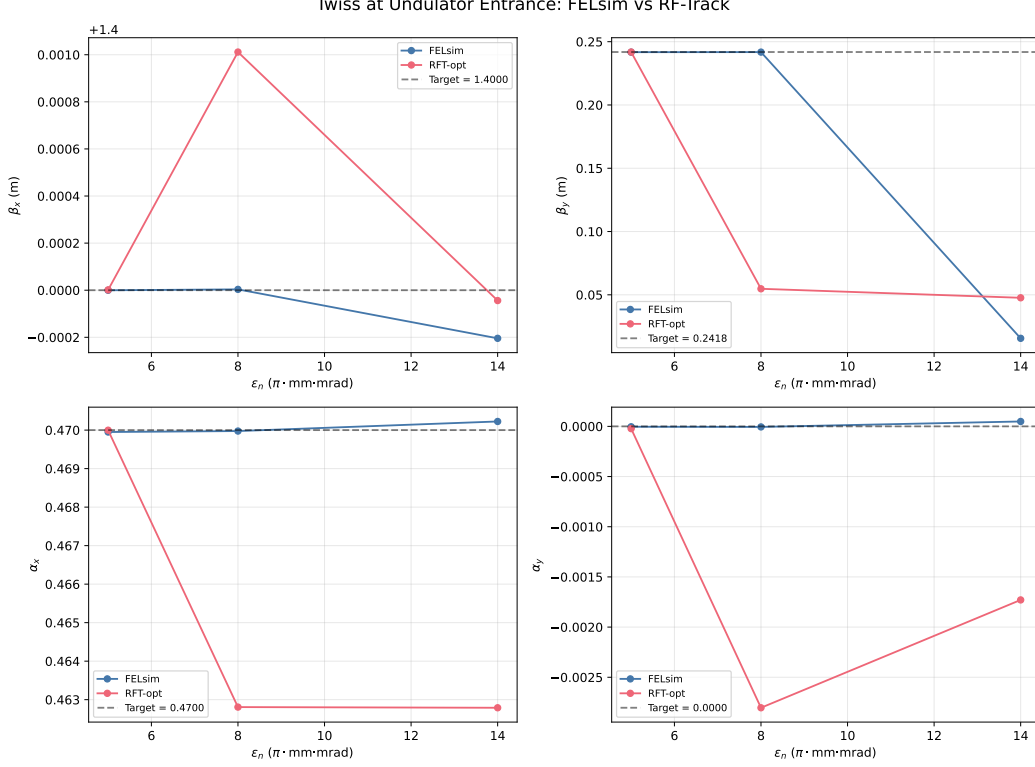


Figure 2: Twiss parameters at the undulator entrance. Targets shown as dashed lines. Both FELsim and RFT-opt match β_x and α_x well across all emittance points.

6 Summary

1. The analytical sector-bend correction (C5) reduces RFT-val MSE from 2552 to 1.99 at $\epsilon_n = 8$, restoring correct dipole body focusing and dispersion in RF-Track.
2. RFT-opt matches all undulator targets closely in x -plane (β_x , α_x) but has a systematic β_y deficit at $\epsilon_n \geq 8$ due to the missing triangle-rule fringe correction ϕ in the thin-lens edge-kick model.
3. At $\epsilon_n = 5$, RFT-opt achieves $\text{MSE} = 2.6 \times 10^{-7}$, $110\times$ better than FELsim (2.85×10^{-5}), suggesting a smoother optimization landscape for particle-based Twiss computation.
4. Prefix caching makes particle-tracking optimisation practical for all tested emittance points.
5. Three additional adapter bugs were fixed during this work: the SBend P/ δ confusion (upstream), the R-matrix unit conversion error, and the edge-kick sign inversion for negative-angle dipoles.

Next steps.

- Implement the triangle-rule fringe correction (ϕ) in the RF-Track thin-lens edge-kick model to resolve the β_y residual.
- Enable space charge in RF-Track to quantify collective effects on the undulator match.

- Compare RF-Track-optimised currents against the COSY FR 1 solution (W4) for a three-code comparison at $\varepsilon_n = 8$.

References

- [1] M. Weinberg, A. Fisher, and R. Li, “Design of the UH Mark V Free Electron Laser,” arXiv:2510.14061v1, Table I.
- [2] A. Latina, *RF-Track Reference Manual*, version 2.5.5, CERN, 2025.